

The following questions are intended to guide you through the most important concepts that you must understand from the lectures and the material provided.

A good exercise would be to study the lectures/course material until you are able to answer them without having to read or search for it in the slides.

Lecture 1: Basic definitions & things, applications, sectors, market

Questions: traditional methods of identity

- which are the traditional methods of verifying the identity of a person (slide 5)
- which are their main problems (slides 6-11)

Question: biometric technologies (slide 13)

- What do they refer to?

Questions: traditional methods vs. biometric technologies (slide 14)

- How do they compare in terms of security, traceability and complexity

The “two samples” aspect is important (slides 16-22)

- Biometrics is about comparing signals (pictures of the face, fingerprint, iris, etc.). When the system compares two biometric signals, it will give an answer about how similar they are. If the two signals are sufficiently similar, then I may conclude that they are from the same person (but NOT who is this person).

Question: Types of biometrics characteristics (slides 25-26)

- Which are the two types of biometrics characteristics and which are the main differences

Question: Attributes of a biometric trait (slide 27)

- Which are the attributes must satisfy a human characteristic to “qualify” as biometric trait

Question: How to choose a biometric trait (slide 29)

- The list of factors is not exhaustive, what it is important to understand here is that there is an interplay of factors, of which accuracy is one of them, but not the only one

Question: Applications of biometrics systems

- Which are the main types of applications (slides 31 -34)
- How do they compare in terms of purpose, cooperativity of the user and control in the acquisition (slide 35)
- Which are the most popular biometrics (slide 36), and others in deployment (37)

Market share and the biometrics revolution (slides 38-41)

Of course, you do not have to study the numbers, just understand tendencies. Watch the recorded lecture for more details

- Before: dominance of public sector. Now: increase of commercial (civil) applications
- Before: dominance of fingerprint+AFIS. Now: increase of modalities like iris, voice, signature
- Before: dominance of (public) identity services. Now: increase of logical access+surveillance

- Before: dominance of North America+Europe. Now: increase of Central/South America+Asia
- Different paradigm in the use of biometrics between “rich” and “poor” countries (with examples in slides 42-46)

Market share and the biometrics revolution (slides 47-54)

- More mobiles than population!
- Plenty of sensors embedded even in low-end devices!
- Acquisition possible in obtrusive and un-obtrusive way (slide 53), which can be for good and for bad (watch the recorded lecture)